

GBA Pro

Custom Handheld Console

Complete Engineering Reference

RP2350 · ILI9341 IPS Display · Authentic GBA Feel
PCB Architecture · KiCad Schematics · Fusion 360 Shell
Power Management · Audio Chain · Firmware Stack
BOM · Fabrication · Assembly · Bring-Up Sequence

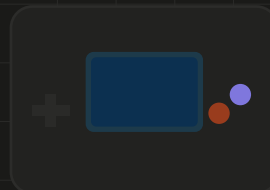


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Chapter 0

Electronics Fundamentals — Everything You Need to Know

Before you touch KiCad, solder a single component, or read a datasheet, you need a solid grasp of the building blocks that every PCB is made of. This chapter is a no-nonsense, practical reference for every fundamental concept used in this project — explained the way a mentor would explain them, not the way a textbook would. If you understand everything in this chapter, you understand every decision made on the GBA Pro PCB.

0.1 — Voltage, Current, and Resistance (Ohm's Law)

These three are the foundation of everything. You cannot design a circuit without them. Think of electricity like water in a pipe:

Electrical Concept	Symbol / Unit	Water Analogy	Plain English
Voltage	V (Volts)	Water pressure	The 'push' that drives electrons through a wire. 3.3V is the pressure this project runs at.
Current	I (Amperes / mA)	Water flow rate	How many electrons are actually moving. The RP2350 draws ~150mA. Your USB port provides 500mA.
Resistance	R (Ohms, Ω)	Pipe diameter	How much a component fights the flow of current. A 10k Ω resistor lets very little through.
Power	P (Watts)	Water horsepower	$P = V \times I$. The heat and energy consumed. At $3.3V \times 0.35A = 1.15W$ — that's warm but fine.

OHM'S LAW — the one equation you must know
 $V = I \times R$. Rearranged: $I = V / R$ and $R = V / I$. Example: you have 3.3V and a 10kOhm pull-up resistor. How much current flows? $I = 3.3V / 10,000\text{Ohm} = 0.33\text{mA}$. That tiny current is enough to hold a logic line HIGH without wasting meaningful power. This is why pull-up resistors are always high values (4.7kOhm-100kOhm).

0.2 — Resistors

A resistor limits current flow. That's it. But that one job makes resistors the most-used component on any PCB. On the GBA Pro, resistors appear in four roles:

Role	Where used in GBA Pro	Why / How it works
Pull-up resistor	All 10 button GPIO pins (R1-R10, 10kOhm)	When no button is pressed, the GPIO pin floats -- it picks up noise and gives random readings. A pull-up resistor connects the pin to 3.3V through 10kOhm, keeping it solidly HIGH (1) when open, and letting it go LOW (0) when the button shorts it to GND.
Current-limiting resistor	LED circuits, USB D+/D- lines (270Ohm)	LEDs will draw infinite current and burn out instantly without a series resistor. The resistor limits current to a safe level. Formula: $R = (V_{\text{supply}} - V_{\text{forward}}) / I_{\text{desired}}$.
Series termination resistor	SPI data lines to MicroSD (100Ohm, R11)	High-speed signals reflect at impedance discontinuities like connectors. A small series resistor (22Ohm-100Ohm) damps the reflection, preventing signal ringing.

Role	Where used in GBA Pro	Why / How it works
Voltage divider	Battery voltage sensing (optional)	Two resistors in series from Vbat to GND. The junction voltage = $V_{bat} \times R2/(R1+R2)$. Scales 4.2V battery down to 3.3V max for ADC reading.

Reading resistor values: On 0402 SMD parts you'll see codes like '103' or '104'. This is mantissa-exponent: $10 \times 10^3 = 10,000\Omega = 10k\Omega$. '104' = $10 \times 10^4 = 100k\Omega$. Modern parts often print the value directly: '10K' or '100R' (R = Ω).

0.3 — Capacitors: The Component That Confuses Everyone

A capacitor stores a small amount of electrical charge. Think of it as a tiny rechargeable battery that charges and discharges in microseconds. This one ability -- storing and releasing charge very quickly -- makes capacitors essential for three completely different jobs. Understanding the difference between these three jobs is the key insight most beginners miss.

Job 1: Decoupling Capacitors (the most important ones on your PCB)

Every IC on this board -- the RP2350, the PCM5102A, the MAX17048 -- draws current in short, sharp bursts when it switches logic states. These bursts happen millions of times per second. Each burst is a tiny voltage dip on the power supply rail. If the dip is large enough, the IC misbehaves -- corrupted data, random resets, or digital noise bleeding into audio.

A **decoupling capacitor** (also called a bypass capacitor) sits directly between the VCC and GND pins of an IC -- physically as close as possible. When the IC demands a burst of current, the capacitor instantly supplies it from its local charge, before the main power supply even notices. The power rail stays stable. No glitches.

DECOUPLING RULE — burn this into your brain

Every IC power pin gets a 100nF (0.1uF) ceramic capacitor from VCC to GND, placed within 0.5mm of the pin. No exceptions. Every. Single. Power. Pin. On the GBA Pro this means C1-C6 (logic ICs) and C9-C12 (audio ICs). If you skip these, your project will work on the bench and fail randomly in the field.

Why 100nF? Ceramic capacitors have a property called self-resonant frequency -- the frequency at which they work best. A 100nF 0402 ceramic resonates at roughly 50-200MHz, which is exactly the frequency range of the switching noise produced by logic ICs running at MHz speeds. It's the right tool for the right job.

Why also add a 10uF bulk cap? The 100nF handles high-frequency noise. But slower, larger current demands -- like the display backlight switching on, or the amp producing a loud sound -- need more charge than 100nF holds. A 10uF capacitor (C7, C8 on the LDO output) handles these mid-frequency events. Together: 10uF + 100nF = full-spectrum power filtering.

Capacitor value	What it filters	Where to place it	GBA Pro usage
100nF (0.1uF)	High-freq switching noise from IC logic (1MHz-200MHz)	Within 0.5mm of every IC VCC pin	C1-C6 (logic), C9-C12 (audio)
10uF	Mid-freq load transients (display, amp, LDO output)	Near LDO output, near amp supply	C7, C8
100uF+	Low-freq bulk energy (motor, large load changes)	Near power input connector	Not needed in this design

Job 2: Coupling Capacitors (DC blocking)

This is the one that confuses people. A coupling capacitor is used in audio circuits to pass AC signals (sound waves) while blocking DC voltage. Here's why that matters:

The TPA2012D2 amplifier in this project outputs an audio signal that sits on top of a DC bias -- roughly half the supply voltage. If you connect the amplifier output directly to a speaker, that DC current flows constantly through the speaker coil, heating it up and potentially destroying it. It also prevents the speaker cone from sitting at its neutral position, distorting the sound.

A **coupling capacitor** (C13, C14 in this project, 10uF each) is placed in series with the audio output line. Capacitors block DC -- they won't pass a steady voltage. But they pass AC freely (the audio signal is AC -- it alternates up and down). The DC is blocked. The audio gets through. The speaker is protected and sounds correct.

COUPLING CAP MENTAL MODEL

Imagine the audio signal as waves on a river, but the river is flowing at 1.65V DC instead of 0V. The coupling capacitor is a dam that stops the river (DC) but lets the waves (AC audio) pass over the top. The speaker only sees the waves -- centred at 0V -- exactly what it needs.

Why 10uF for audio coupling? The capacitor and speaker coil form a high-pass filter. The cutoff frequency is $f = 1 / (2\pi \times R \times C)$. A 10uF cap with an 8Ohm speaker gives $f = 2\text{Hz}$ -- well below the 20Hz lower limit of human hearing. All audible bass gets through. If you used 100nF instead, the cutoff would be 200Hz -- you'd lose all bass. Bigger cap = lower bass cutoff.

Job 3: Filter Capacitors (power supply smoothing)

In power supplies, capacitors smooth out ripple -- the small AC variation left over after rectification or switching. In this project the LDO (XC6220A) is already very clean, so explicit filter caps aren't critical, but the 10uF on the output still improves transient response and reduces output noise.

0.4 — Pull-Up and Pull-Down Resistors

A floating input pin is an engineer's nightmare. When nothing is driving a GPIO pin -- no button pressed, no IC actively pulling it -- the pin picks up electromagnetic interference from nearby traces and reads as random noise. Pull resistors fix this.

Type	Connection	Idle state	Active state	Used for
Pull-up	Pin -> R -> VCC (3.3V)	HIGH (1)	Goes LOW when button shorts to GND	All GBA Pro buttons. Default in most microcontrollers.
Pull-down	Pin -> R -> GND	LOW (0)	Goes HIGH when input connects to VCC	Less common. Used when active-high logic is needed.
Internal pull-up	Inside the RP2350 chip	HIGH (1)	Software-configurable per pin	RP2350 has built-in 50kOhm pull-ups. External 10kOhm used here for stronger, more noise-immune drive.

Why 10kOhm and not 1kOhm or 100kOhm? Lower resistance = stronger pull = more noise immunity = more current wasted when button is pressed ($3.3\text{V} / 1\text{kOhm} = 3.3\text{mA}$ per button -- 10 buttons = 33mA wasted). Higher resistance = weaker pull = less current waste = more susceptible to noise and slower signal edges. 10kOhm is the standard compromise: 0.33mA per button, excellent noise immunity, total 3.3mA for all 10 buttons.

0.5 — Voltage Regulators and LDOs

A battery outputs variable voltage -- a LiPo ranges from 4.2V (full) down to 3.0V (empty). ICs need a stable, precise voltage. A voltage regulator sits between the battery and the circuit, absorbing the variation and outputting a clean, fixed voltage.

The **XC6220A** in this project is an LDO -- Low Drop-Out regulator. 'Low drop-out' means it can maintain its output voltage even when the input is only slightly higher. The XC6220A needs just 150mV of headroom: it will output clean 3.3V as long as the battery is above 3.45V. This means the device keeps working reliably until the battery is genuinely flat.

LDO vs Switching Regulator — why LDO here?

A switching regulator (buck/boost) is more efficient but generates electrical noise. An LDO is less efficient (wastes energy as heat: $P_{\text{loss}} = (V_{\text{in}} - V_{\text{out}}) \times I = \sim 0.1\text{V} \times 0.3\text{A} = 30\text{mW}$) but produces a very clean output. For audio circuits, clean power is worth the efficiency cost. At 30mW heat loss on a 1.15W system, the efficiency is 97%+ -- totally acceptable.

0.6 — Digital Communication Protocols: SPI and I2C

ICs on a PCB talk to each other using standardised protocols. The GBA Pro uses two: SPI for high-speed (display, SD card) and I2C for low-speed (fuel gauge). Understanding these at a basic level helps you read datasheets and debug problems.

SPI — Serial Peripheral Interface

SPI is the fast lane. The RP2350 can run SPI at 62.5MHz -- that's 62.5 million bits per second. This speed is needed to push a full 320x240 frame to the display fast enough for smooth animation. SPI uses 4 wires:

Wire	Full Name	Direction	Function
SCK / CLK	Serial Clock	Master -> Slave	The heartbeat. Both sides synchronise on every rising edge. Faster clock = more data per second.
MOSI / TX	Master Out Slave In	Master -> Slave	The data going from RP2350 to the display or SD card.
MISO / RX	Master In Slave Out	Slave -> Master	Data coming back from the SD card to RP2350. The display is write-only so MISO is NC for it.
CS / CSn	Chip Select (active LOW)	Master -> Slave	Tells the specific IC 'I'm talking to you now.' Pull LOW to select, HIGH to deselect. Each device needs its own CS line.

DC (Data/Command) pin on ILI9341: The display controller needs to know if the bytes coming in are commands (change display settings, set cursor position) or pixel data. The DC pin solves this: pull it LOW before sending a command byte, HIGH before sending pixel data. This is specific to display controllers -- it's not a standard SPI pin.

I2C — Inter-Integrated Circuit

I2C is the slow lane -- up to 400kHz (0.4MHz) -- but it only needs 2 wires to connect multiple devices. The MAX17048 fuel gauge uses I2C. Both wires require pull-up resistors (4.7kOhm in this project) because I2C devices communicate by pulling lines LOW and releasing them -- without pull-ups, released lines float instead of returning to HIGH.

Wire	Full Name	Function
SDA	Serial Data	Carries the actual data -- addresses and values. Bidirectional: both master and slave drive it.

Wire	Full Name	Function
SCL	Serial Clock	The timing signal driven by the master (RP2350). All slaves listen and respond on SDA.

I2C addresses: Every I2C device has a 7-bit address baked in at the factory. The MAX17048 is always at address 0x36. When the RP2350 wants to read the battery percentage, it sends 0x36 on SDA -- every I2C device hears it, but only the MAX17048 (whose address matches) responds. This is how 2 wires can talk to many different chips.

0.7 — Digital Audio: I2S

I2S (Inter-IC Sound) is the standard protocol for sending digital audio between chips. The RP2350 sends audio data over I2S to the PCM5102A DAC, which converts the digital stream into an analogue waveform. Three wires:

Wire	Name	Function
BCK	Bit Clock	Clocks each individual audio bit. Frequency = sample_rate x bit_depth x channels. At 44100Hz, 16-bit, stereo: BCK = 44100 x 16 x 2 = 1.41MHz.
LRCK / WS	Left-Right Clock / Word Select	Switches between LEFT and RIGHT audio channel once per sample. Frequency = sample rate (44100Hz or 48000Hz).
DATA / DIN	Serial Data	The actual audio samples -- a stream of bits representing the amplitude of each audio sample.

The RP2350 has no hardware I2S peripheral, so the GBA Pro uses PIO (Programmable IO) -- small state machines inside the RP2350 that you program in a simple assembly language to implement any custom protocol. PIO is one of the RP2350's most powerful features.

0.8 — Ground Planes and Signal Return Paths

Ground is not just 'the negative terminal.' Ground is the return path for every single current in your circuit. Every milliamp that flows through the RP2350, the display, the amp -- every one of those currents has to flow back to the source through the ground network. If the ground path is long, thin, or shared with another current, you get ground bounce -- a tiny but real voltage difference between two 'ground' points that causes mysterious failures.

THE GROUND PLANE RULE

On any 2-layer PCB: fill the entire bottom copper layer with GND. This is called a ground plane. It gives every current the shortest possible return path and dramatically reduces radiated noise. In KiCad, use Edit > Fill All Zones after assigning the bottom copper zone to GND. Without a ground plane, your high-speed SPI traces will radiate like antennas.

0.9 — PCB Trace Width and Current Capacity

A PCB trace is a wire -- it has resistance, and carrying current through resistance generates heat. A trace that's too thin will overheat, delaminate, and eventually open-circuit. The IPC-2221 standard gives guidance, but these rules cover 95% of cases:

Current	Minimum trace width (external layer)	Use case in GBA Pro
< 1mA	0.1mm (4mil)	I2C signals, GPIO signals -- any logic that barely carries current
< 100mA	0.2mm (8mil)	SPI lines, audio signals, button lines
< 300mA	0.4mm (16mil)	3.3V logic power rail branches

Current	Minimum trace width (external layer)	Use case in GBA Pro
< 500mA	0.6mm (24mil)	Main 3.3V power rail from LDO to Pico 2
< 1000mA	1.0mm (40mil)	Battery input, USB VBUS to TP4056 charger
> 1A	1.5mm+ (60mil+)	Not used in this design -- all currents are under 1A

0.10 — The RP2350 Pico 2's Decoupling Requirements

The Pico 2 module handles its own internal decoupling for the RP2350 core. But the VSYS pin (your 3.3V input) still needs external decoupling at the board level. The Pico 2 datasheet recommends placing a 10uF + 100nF capacitor pair on VSYS close to the module. This is already included in the GBA Pro BOM as C7 and C8.

0.11 — How to Read a Datasheet

A datasheet is the manufacturer's complete specification for a component. They look intimidating but follow a standard structure once you know what to look for:

Section	What's in it	What to look for
Features / Description	Marketing summary of what the chip does	Confirm it's the right chip. Check key specs: voltage range, current, interface type.
Pin Configuration / Pinout	Diagram showing every pin by number and name	This is critical. Cross-check every pin against your schematic. One wrong connection destroys the chip.
Absolute Maximum Ratings	The limits the chip will survive -- but only briefly	Never operate at these. They're the 'one second before it dies' numbers. Always stay in the Recommended Operating Conditions instead.
Recommended Operating Conditions	Voltage, temperature, current ranges for normal use	Your design must stay inside these at all times. Check min/max Vcc, logic levels, temperature range.
Electrical Characteristics	Precise spec of every pin's behaviour	Check: input high/low voltage thresholds (is 3.3V logic compatible?), output drive current, quiescent current.
Application Circuit / Typical Application	The manufacturer's recommended schematic	Start here. Copy this circuit exactly for your first prototype. Add only what you specifically need to change.
Timing Diagrams	Waveform diagrams showing protocol timing	Needed for SPI/I2S troubleshooting. Shows setup/hold times and clock polarity.
Package / Footprint	Physical dimensions of the chip	Download the footprint if KiCad doesn't have it, or verify the dimensions match the KiCad footprint exactly.

0.12 — Common Beginner Failures and How to Avoid Them

Failure	Why it happens	How to prevent it
IC gets hot and dies immediately	Power pins reversed, or voltage too high. Even 0.3V over the absolute max can kill a chip in milliseconds.	Triple-check VCC and GND pin orientation before powering on. Use a bench supply with current limiting set to 100mA for first power-on.
Circuit works on bench, fails randomly later	Missing decoupling caps. Power rail dips during load changes causing glitches.	Add 100nF decoupling cap to every IC power pin. This is the #1 cause of random, unreproducible bugs.

Failure	Why it happens	How to prevent it
Display shows garbage or nothing	SPI DC pin wrong, or CS not pulled low during transaction, or clock polarity mismatch (CPOL/CPHA).	Check DC pin state before each command/data write. Verify SPI mode -- ILI9341 uses Mode 0 (CPOL=0, CPHA=0).
Audio has hiss or buzz	Ground loop or shared return path between digital and analogue ground. Or no coupling cap on output.	Keep analogue ground return traces physically separate from digital. Add output coupling caps (C13, C14).
Buttons register wrong or randomly	No pull-up resistors, or pull-up too weak (100kOhm+) making pin susceptible to noise.	Use 10kOhm pull-ups on every button pin. Confirm in firmware with GPIO_PULL_UP flag.
SD card fails to initialise	SPI speed too high for card init. SD cards must init at 400kHz then can go faster.	Initialise SD at 400kHz. Only increase speed after successful init. Check series resistor on SPI lines.
Board works but drains battery in hours	A pull-up or indicator LED drawing constant current. Or IC in wrong power mode.	Calculate total quiescent current: sum all Iq values from datasheets. Measure actual current with multimeter in series with battery.
Solder bridge causing short	Too much solder paste on fine-pitch QFN or TSSOP pads.	Use a stencil for paste application. Inspect under magnification before powering on. Use solder wick + flux to remove bridges.

THE GOLDEN RULE OF BRING-UP

Never power a new board at full voltage immediately. Use a bench power supply with current limiting. Set the current limit to 50mA and slowly raise voltage from 0V. If current spikes before voltage reaches 3.3V, there is a short -- stop immediately, find it, fix it. A 5-second short at full current can destroy every IC on the board. This one habit will save you hundreds of dollars.

Chapter 1

Project Philosophy & Concept

This document is the complete engineering reference for the GBA Pro — a scratch-built, RP2350-powered handheld game console that pays faithful homage to Nintendo's Game Boy Advance (AGB-001, released 2001). Every decision made in this project is driven by one principle: **resurrect the feel of the original, powered by modern silicon.**

The Core Philosophy

The GBA Pro is not an upgrade of the original. It is a resurrection. The soul of this project lives in the D-pad — that satisfying cruciform, 32mm spanning, conductive-rubber-domed input that Nintendo's engineers got exactly right in 2001. The AB buttons, flat and precise. The shoulder buttons that snap with authority. These are not to be changed.

What changes is everything the player never sees: the processor, the display backlight, the charging port, the power management. The outside is sacred. The inside is modern.

DESIGN RULE #1

Never change anything the player touches. Rubber dome actuation, D-pad geometry, shell corner radius — all preserved exactly to AGB-001 spec. Modernisation happens invisibly.

What This Device Does

Feature	Detail
Plays	Game Boy Advance ROMs from MicroSD — native resolution 240x160, displayed pixel-perfect on a 320x240 IPS panel.
Charges	Via USB-C, with real battery percentage readout on the UI.
Sounds	Better than the original — PCM5102A I2S DAC + Class-D amp versus the GBA's crude PWM audio.
Boots	Instantly. No Linux. No boot animation. Power on — game loads.
Lasts	7 to 8 hours on a 2500mAh LiPo. The original GBA used 2x AA batteries for ~15 hours, but this is rechargeable.

What This Device Is Not

This is not a multi-emulator handheld. It does not run N64, PS1, or SNES. Those are separate projects with separate hardware requirements. This device is dedicated to GBA — and that focus is a feature, not a limitation. The RP2350 is sized exactly right for GBA emulation, runs cool, draws little power, and needs no operating system.

WHY RP2350 NOT RASPBERRY PI CM4

The CM4 runs Linux, draws 500mA+, takes 15 seconds to boot, and costs \$35. The RP2350 runs bare metal, draws ~150mA, boots in milliseconds, and costs \$5. For GBA-only, the CM4 is a V8 engine in a bicycle. The RP2350 is the right tool.

Chapter 2

Architecture Overview

The GBA Pro is a single-board design. One custom PCB carries every component — processor, power management, audio chain, input logic, display interface, and storage. There is no co-processor, no secondary board, no daughtercard. Everything fits on a 130 x 70mm 2-layer PCB.

System Block Diagram

The following table maps every major block, its function, and its interface to the RP2350:

Block	Component	Interface	RP2350 Pins
Processor	Raspberry Pi Pico 2 (RP2350)	—	All GPIO via stamp module
Display	ILI9341 2.8" IPS 320x240	SPI0 @ 62.5MHz	GP2(CLK) GP3(TX) GP5(CS) GP6(DC) GP7(RST)
MicroSD	Molex 104031-0811	SPI1 @ 25MHz	GP10(CLK) GP11(TX) GP12(RX) GP13(CS)
Audio DAC	PCM5102A I2S	PIO I2S	GP16(BCK) GP17(LRCK) GP18(DATA)
Audio Amp	TPA2012D2 Class-D	Analogue from DAC	—
Speaker	28mm 8-ohm 0.5W	Amp output	—
3.5mm Jack	PJ-320A SMD	Amp output (switched)	GP19(detect)
Charger	TP4056	USB-C VBUS in	—
LDO	XC6220A 3.3V 700mA	LiPo → 3.3V	VSYS pin
Fuel Gauge	MAX17048	I2C1	GP20(SDA) GP21(SCL)
Buttons	Rubber dome + carbon pads	GPIO pull-up	GP0–GP9 (10 inputs)
L/R Shoulder	Panasonic EVQP7C01P tact	GPIO pull-up	GP8 GP9
USB-C Port	GCT USB4135-GF-A	TP4056 charge + GP USB data	USB DP/DM
LiPo Battery	2500mAh 3.7V 100x55x5mm	TP4056 BAT pin	—

Power Architecture

The power chain is linear and simple by design. USB-C 5V feeds the TP4056, which charges the LiPo at 1A. The LiPo output (3.7V nominal, 3.0V–4.2V range) feeds the XC6220 LDO, which regulates to a stable 3.3V. Everything — the Pico 2, display, DAC, amp, fuel gauge — runs from this single 3.3V rail.

Stage	Component	Vin	Vout	Max Current
USB Input	USB-C port	5V (USB)	5V	500mA
Charge	TP4056	5V	4.2V (LiPo)	1000mA
Battery	2500mAh LiPo	—	3.0–4.2V	2500mAh capacity
Regulation	XC6220A LDO	3.0–4.2V	3.3V ±1%	700mA
Logic load	Pico 2 + display + ICs	3.3V	—	~350mA peak

BATTERY LIFE ESTIMATE

At 150mA idle (Pico2 + display backlight at 50%) the 2500mAh cell gives ~16 hours. At full load (350mA — backlight 100%, audio loud, heavy emulation) expect 7 hours. Typical gameplay at moderate brightness: 9–10 hours.

Signal Domains

The PCB has three signal domains that must be kept physically separated during layout: **Digital high-speed** (SPI display and SD), **Digital low-speed** (GPIO buttons, I2C fuel gauge), and **Analogue audio** (I2S DAC output, amp input). Mixing these causes noise. The layout chapter defines keep-out zones for each domain.

Chapter 3

Phase 1 — Bill of Materials

Every component required for one complete prototype unit. Quantities are for a single build. Order at least 2x of all SMD passives (resistors, capacitors) to account for reflow losses. Total estimated cost: ~\$85 USD for components + PCB fabrication + one printed shell.

BRAIN

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
U1	Raspberry Pi Pico 2	SC0918	raspberrypi.com / Digikey	1	\$5.00	RP2350, stamp module, boot via SD

DISPLAY

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
LCD1	2.8" IPS TFT 320x240	Adafruit 1770 equiv.	Adafruit / Aliexpress	1	\$14.00	ILI9341 driver, SPI, 3.3V logic

POWER MANAGEMENT

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
U2	TP4056 LiPo charger	TP4056	LCSC / Aliexpress	1	\$0.50	1A charge, built-in OVP/UVP
U3	XC6220A LDO 3.3V	XC6220A331MR-G	LCSC	1	\$0.60	700mA, ultra-low dropout 150mV
U4	MAX17048 fuel gauge	MAX17048G+T	Digikey / Mouser	1	\$2.00	I2C, 1% SOC accuracy
BT1	LiPo 2500mAh 3.7V	Generic 102555	Adafruit / Aliexpress	1	\$10.00	100x55x5mm, JST-PH 2.0
J1	USB-C receptacle	USB4135-GF-A	Digikey	1	\$1.00	Mid-mount, 2mm height
SW1	Power slide switch	SS-12D00-G3	LCSC	1	\$0.50	DPDT, 3-pin, panel mount
C1–C6	100nF decoupling caps	CC0402KRX7R7BB104	LCSC	10	\$0.02	0402, X7R, one per power pin
C7–C8	10uF bulk caps	GRM188R61A106KE69	LCSC	4	\$0.15	0402, LDO in/out

INPUT

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
SW2–SW9	Rubber dome sheet (D-pad+AB+Sel+Sta)	GBA-spec dome sheet	Handheld Legend / Ali	1	\$8.00	160g actuation, 0.5mm travel
SW10–SW11	Shoulder tact switch	EVQP7C01P	Digikey	2	\$0.40	Panasonic, 100g, 0.8mm travel
R1–R10	10k pull-up resistors	RC0402FR-0710KL	LCSC	12	\$0.01	0402, one per button GPIO

AUDIO

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
U5	PCM5102A I2S DAC	PCM5102APWR	Digikey / LCSC	1	\$3.00	112dB SNR, TSSOP-20
U6	TPA2012D2 Class-D amp	TPA2012D2RTJR	Digikey	1	\$4.00	2.5W/ch, filterless, QFN-16
SP1	Speaker 28mm 8-ohm	28mm round driver	Mouser / Aliexpress	1	\$3.00	0.5W, 4mm depth, mono
J2	3.5mm headphone jack	PJ-320A	LCSC	1	\$0.50	SMD, auto-mute on insert
C9–C12	100nF audio decoupling	CC0402KRX7R7BB104	LCSC	4	\$0.02	0402, DAC power pins
C13–C14	10uF output coupling	GRM188R61A106KE69	LCSC	2	\$0.15	0402, amp output

STORAGE

Ref	Component	Part Number	Supplier	Qty	Unit \$	Notes
J3	MicroSD push-push slot	104031-0811	Digikey / Mouser	1	\$2.00	Molex, 8-pin, bottom contact
R11	100 ohm series resistor	RC0402FR-07100RL	LCSC	1	\$0.01	0402, SPI data lines EMI

FABRICATION

Item	Specification	Supplier	Qty	Est. Cost	Notes
PCB	2-layer, ENIG, 130x70mm, 1.6mm FR4	JLCPCB	5 boards	\$18.00	Order 5, build 1, keep 4 spares
Shell front	PETG, 0.15mm layer, 40% infill	Self-print / service	1	\$4.00	144x82mm footprint
Shell back	PETG, 0.15mm layer, 40% infill	Self-print / service	1	\$4.00	Includes battery bay
M2 inserts	Brass heat-set, 3mm length	Amazon / Aliexpress	10	\$3.00	4 shell + 4 PCB standoffs
M2 screws	Stainless, 5mm length	Amazon / Aliexpress	12	\$1.00	Phillips pan head
Solder paste	Sn63Pb37 0.5mm syringe	Chipquik / Amazon	1	\$8.00	For SMD hand soldering

TOTAL (1 unit) ~\$85.00

Chapter 4

Phase 2 — KiCad PCB Design

This chapter walks through every stage of the PCB design — from project setup in KiCad 7+ through schematic capture, library setup, layout, design rule check, and Gerber export. Follow each section in order. Do not move to PCB layout until the schematic DRC passes clean.

4.1 KiCad Project Setup

Create a new KiCad project. Name it **gba_pro_v1**. Set the following preferences before drawing anything:

Setting	Value	Why
Schematic grid	50 mil (1.27mm)	Standard schematic grid — all symbols snap correctly
PCB grid (coarse)	0.5mm	Component placement grid
PCB grid (fine)	0.1mm	Via and pad fine adjustment
Schematic page size	A3	Enough room for all blocks without crowding
Net name length	Keep short (<12 chars)	Long net names crowd the schematic
Title block	GBA Pro v1.0 / Rev A	Appears on every printout
Minimum trace width	0.2mm	2-layer board safe minimum
Minimum clearance	0.2mm	JLCPCB standard capability
ENIG finish	Yes — specify in fab notes	Required for rubber dome carbon pad contacts

4.2 Symbol Libraries

Add these libraries before placing any symbols. In KiCad: Preferences → Manage Symbol Libraries:

Library	Source	Contains
Raspberry_Pi	KiCad official	Pico 2 / RP2350 stamp module
Device	KiCad official	Resistors, capacitors, generic passives
Connector	KiCad official	USB-C, MicroSD, JST connectors
Analog_DAC	KiCad official	PCM5102A (or add manually)
Texas_Instruments	KiCad official	TPA2012, MAX17048
Linear_Technology	KiCad official	TP4056, XC6220 (or add manually)
Custom_GBA_Pro	Create new	Any parts not in official libs

NOTE: For parts not in official libraries (TP4056, XC6220, ILI9341), download KiCad symbols from [SnapEDA.com](https://snapeda.com) or create them manually. Creating manually takes 10 minutes per part — just map pin numbers to a rectangle.

4.3 Schematic — Power Sheet

Create the power section on its own schematic sheet (multi-sheet design keeps things clean). This sheet contains: USB-C connector → TP4056 → LiPo battery → XC6220 LDO → 3V3 power symbol. Also place MAX17048 on this sheet as it is purely power-domain.

Draw in this exact order:

Step	Action	Detail
1	Place +3V3 and GND power symbols	These will be your reference rails throughout
2	Place USB-C symbol (J1)	Wire VBUS to TP4056 VIN, GND to GND rail
3	Add 5.1k CC resistors (R_CC1, R_CC2)	Both CC pins pull to GND via 5.1k — this signals USB-C power source
4	Place TP4056 (U2)	VIN from USB VBUS, BAT to battery net, GND, PROG pin with 1.2k resistor to GND (sets 1A charge)
5	Place JST-PH 2.0 battery connector	BAT+ from TP4056 BAT, BAT- to GND
6	Place XC6220A (U3)	VIN from BAT net, VOUT to +3V3 rail, add 1uF input and 1uF output cap to GND
7	Place MAX17048 (U4)	VCC to +3V3, GND, SDA to I2C_SDA net, SCL to I2C_SCL net, CELL+ to BAT net, CELL- to GND
8	Add all decoupling caps	100nF cap on every VCC pin of every IC, placed in schematic next to pin
9	Add power switch SW1	DPDT switch in series between LiPo BAT+ and LDO VIN
10	Run ERC	Fix all errors before moving to next sheet

4.4 Schematic — Processor Sheet

Place the Raspberry Pi Pico 2 symbol. This is the largest symbol on the schematic. Pin assignments are non-negotiable — the SPI0 peripheral is hardwired to specific GPIO on the RP2350. You cannot move the display to arbitrary pins.

SPI0 (Display): These pins are fixed by the RP2350 SPI0 peripheral:

Signal	GPIO	Pico 2 Pin	Connect to	Note
SPI0_CLK	GP2	Pin 4	ILI9341 SCK	50MHz max on this pin
SPI0_TX (MOSI)	GP3	Pin 5	ILI9341 SDA/MOSI	Data to display
SPI0_CSn	GP5	Pin 7	ILI9341 CS	Active low chip select
DC (Data/Command)	GP6	Pin 9	ILI9341 DC/RS	High=data, Low=command
RST (Reset)	GP7	Pin 10	ILI9341 RESET	Active low reset
Backlight PWM	GP8	Pin 11	ILI9341 LED/BL via transistor	PWM brightness control

SPI1 (MicroSD):

Signal	GPIO	Pico 2 Pin	Connect to	Note
SPI1_CLK	GP10	Pin 14	SD CLK	25MHz for SD
SPI1_TX	GP11	Pin 15	SD CMD/MOSI	

Signal	GPIO	Pico 2 Pin	Connect to	Note
SPI1_RX	GP12	Pin 16	SD DAT0/MISO	
SPI1_CS _n	GP13	Pin 17	SD CS	Active low
SD_DET	GP14	Pin 19	SD Card Detect switch	Pull-up, active low when card inserted

PIO I2S (Audio DAC):

Signal	GPIO	Pico 2 Pin	Connect to	Note
I2S_BCK	GP16	Pin 21	PCM5102A BCK	Bit clock
I2S_LRCK	GP17	Pin 22	PCM5102A LRCK	Left/right word clock
I2S_DATA	GP18	Pin 24	PCM5102A DIN	Serial audio data

I2C1 (Fuel Gauge):

Signal	GPIO	Pico 2 Pin	Connect to	Note
I2C1_SDA	GP20	Pin 26	MAX17048 SDA	4.7k pull-up to 3.3V
I2C1_SCL	GP21	Pin 27	MAX17048 SCL	4.7k pull-up to 3.3V

GPIO Buttons:

Signal	GPIO	Pico 2 Pin	Connected Button	Pull-up
BTN_RIGHT	GP0	Pin 1	D-pad Right	10k to 3.3V
BTN_LEFT	GP1	Pin 2	D-pad Left	10k to 3.3V
BTN_UP	GP2 — CONFLICT	—	— USE GP22 INSTEAD	Reassign if SPI0 conflict
BTN_UP	GP22	Pin 29	D-pad Up	10k to 3.3V
BTN_DOWN	GP26	Pin 31	D-pad Down	10k to 3.3V
BTN_A	GP27	Pin 32	A button	10k to 3.3V
BTN_B	GP28	Pin 34	B button	10k to 3.3V
BTN_SEL	GP15	Pin 20	Select	10k to 3.3V
BTN_STA	GP19	Pin 25	Start	10k to 3.3V
BTN_L	GP8	Pin 11	L shoulder	10k to 3.3V — share with BL PWM or reassign
BTN_R	GP9	Pin 12	R shoulder	10k to 3.3V

CRITICAL: GP2 is used for SPI0_CLK (display). Do NOT also use it for D-pad Up. Use GP22 for D-pad Up instead. Double-check every GPIO assignment against the RP2350 datasheet peripheral mapping table before routing the PCB.

4.5 Schematic — Audio Sheet

Place the audio chain on a dedicated sheet. Signal flow: RP2350 I2S → PCM5102A → TPA2012D2 → Speaker + Headphone jack.

Step	Action
1	Place PCM5102A (U5). Wire BCK, LRCK, DIN from RP2350 I2S nets. VCC to +3V3. GND and AGND both to GND (short them at this IC only). FLT, DEMP, XSMT pins: FLT to GND (normal rolloff), DEMP to GND (no de-emphasis), XSMT to +3V3 via 100k (unmuted).
2	PCM5102A OUTR and OUTL are analogue voltage outputs. Add 100nF DC-blocking cap on each output line before the amp input.
3	Place TPA2012D2 (U6). Wire IN+ and IN- for one channel from PCM5102A output (it's a bridge amp — both + and - inputs). VDD to +3V3. GND. SDZ (shutdown) to +3V3 via 100k (always on). GAIN pins: wire to GND for 12dB gain (suitable for 8-ohm speaker at 3.3V).
4	TPA2012D2 outputs are differential (OUT+ and OUT-). Wire both to speaker pads. Do NOT add a coupling cap — Class-D output is already DC-coupled differentially.
5	Place PJ-320A headphone jack (J2). Wire TRS tip/ring to PCM5102A OUTL/OUTR via 33-ohm series resistors (headphone protection). Wire sleeve to GND. Wire the normally-closed switch pin to a GPIO (GP19) via 10k pull-up — this signal goes LOW when headphones inserted, so firmware can mute the amp.

4.6 PCB Layout — Layer Stack

This is a 2-layer board. Both layers carry signals. No dedicated ground plane — but we manage this deliberately:

Layer	Name	Content
F.Cu (Top)	Signal + power	All SMD components on top. Display, Pico 2, ICs, passives. Main power traces on this layer.
B.Cu (Bottom)	Signal + ground fill	Ground fill flood on entire bottom layer. Button through-hole pads. MicroSD and headphone jack if bottom-mount.
F.Silks	Silkscreen top	Component reference designators, polarity marks, connector labels
F.Paste	Solder paste	Auto-generated from pads. Needed if using stencil (recommended).
Edge.Cuts	Board outline	130 x 70mm rectangle with 1mm corner radii

NOTE: Add a ground fill flood on B.Cu after placing all traces. In KiCad PCB editor: Add → Add Filled Zone → select B.Cu → Net = GND → flood. This gives you a pseudo-ground-plane on a 2-layer board which dramatically reduces EMI from the SPI display lines.

4.7 Component Placement Rules

Component	Placement Rule
Pico 2 module	Centre-right of board. USB port of Pico 2 faces board edge (not used — your USB-C is separate).
ILI9341 display connector	Top-left region. FPC/ribbon connects to display above, exits toward top edge.
TP4056 charger	Bottom-left corner, close to USB-C port. Short VBUS trace = less voltage drop.
XC6220 LDO	Beside TP4056. Short trace from TP4056 BAT to LDO VIN.
PCM5102A DAC	Far right side, away from SPI clock lines. Analogue quiet zone.
TPA2012D2 amp	Beside DAC, close to speaker connector. Short speaker traces.
MicroSD slot	Right edge of board, card slots OUT toward board edge for user access.
USB-C port	Bottom edge, centred. Mid-mount part sits flush with board bottom.
Headphone jack	Bottom edge, right side.

Component	Placement Rule
Button pads	Distributed across board matching shell button well positions exactly.
Battery connector	Bottom-right, JST-PH 2.0 pointing down.
Decoupling caps	Within 1mm of the power pin they decouple. This is non-negotiable.

4.8 Rubber Dome Contact Pad Design

This is the single most important layout detail for authentic GBA feel. The carbon-coated rubber dome shorts two interdigitated PCB pads when pressed. Getting the pad geometry wrong means the button either doesn't register or double-registers.

Parameter	Value	Notes
Pad pattern	Interdigitated fingers	Two interlocking combs, not two separate circles
Total pad diameter	7.0mm	Dome sits over this area
Finger width	0.4mm	ENIG gold finish required — not HASL
Finger gap	0.3mm	Minimum gap for reliable open-circuit when unpressed
Finger count per side	5 fingers each	10 total fingers in the pattern
D-pad pad centres	Match rubber dome sheet spec	Measure your sourced dome sheet first, then lay out pads
AB pad centres	10.0mm diameter circle	A and B pads 12mm apart centre-to-centre
Start/Select pads	6.0mm diameter variant	Smaller dome, smaller pad
Surface finish	ENIG (Electroless Nickel Immersion Gold)	Specify in JLCPCB order — costs ~\$8 extra but critical

CRITICAL: Order your rubber dome sheet **BEFORE** finalising pad layout. Measure the actual dome contact area with calipers, then design your pads to match. Different suppliers have slightly different dome sizes. Your PCB pads must match your actual domes.

4.9 Trace Width Guidelines

Net / Signal	Min Width	Recommended Width	Reason
GND fill	N/A	Flood fill B.Cu	Ground return for all signals
3.3V power rail	0.5mm	0.8mm	Carries up to 350mA peak
VBAT (LiPo to LDO)	0.5mm	1.0mm	Carries charge current up to 1A
USB VBUS (5V)	0.5mm	1.0mm	1A charge current
SPI0 display signals	0.2mm	0.25mm	High-speed — keep short and direct
SPI1 SD signals	0.2mm	0.25mm	25MHz — short traces
I2S audio signals	0.2mm	0.2mm	Keep away from SPI lines
I2C signals	0.2mm	0.2mm	Low speed, not critical

Net / Signal	Min Width	Recommended Width	Reason
GPIO button signals	0.2mm	0.2mm	Low speed, not critical
Speaker output	0.3mm	0.5mm	Carries audio current to speaker

4.10 Design Rule Check & Gerber Export

Step	Action
1 — DRC	Run Inspect → Design Rules Checker. Fix ALL errors. Warnings about courtyard overlap are OK if intentional. Electrical errors are never OK.
2 — Ground flood	Add filled zone on B.Cu (GND net). Refill. Check all GND pads are connected to flood.
3 — 3D view	Use View → 3D Viewer to check for component collisions and verify connector orientations.
4 — Gerber export	File → Fabrication Outputs → Gerbers. Select: F.Cu, B.Cu, F.Silks, F.Mask, B.Mask, Edge.Cuts. Output to /gerbers folder.
5 — Drill files	File → Fabrication Outputs → Drill Files. Generate Excellon drill file. Include in /gerbers folder.
6 — Gerber review	Open gerbers in KiCad Gerber Viewer or upload to JLCPCB instant viewer. Check every layer visually.
7 — BOM export	File → Fabrication Outputs → BOM. Match against Phase 1 BOM to confirm all parts placed.
8 — Upload to JLCPCB	Zip the /gerbers folder. Upload to jlcpcb.com. Select: 2-layer, FR4, 1.6mm, ENIG, green solder mask, 5 pieces.

Chapter 5

Phase 3 — Fusion 360 Enclosure Design

The GBA Pro shell is designed from scratch in Fusion 360, using the original AGB-001 measurements as a strict reference. Every external dimension matches the original. The interior is redesigned to accommodate the custom PCB, larger battery, and USB-C port. This chapter gives you the exact modelling workflow and all critical dimensions.

5.1 AGB-001 Master Dimensions

These measurements are your sacred reference. Do not deviate from any external dimension:

Dimension	AGB-001 Original	GBA Pro Target	Notes
Overall width	144.5mm	144.0mm	-0.5mm for tighter fitment tolerance
Overall height	82.0mm	82.0mm	Exact match
Overall depth	24.3mm	25.0mm	+0.7mm — accommodates USB-C mid-mount
Top corner radius (ext)	12.0mm	12.0mm	Exact match — characteristic GBA curve
Bottom corner radius (ext)	8.0mm	8.0mm	Exact match
Shell wall thickness	1.8mm	2.0mm	Slightly thicker for FDM strength
Display window width	61.6mm	62.0mm	+0.4mm for ILI9341 bezel clearance
Display window height	41.4mm	42.0mm	+0.6mm for ILI9341 bezel clearance
Display window from top	14.0mm	14.0mm	Exact match
Display window from left	41.2mm	41.0mm	Centred horizontally
D-pad centre from left	19.0mm	19.0mm	Exact match
D-pad centre from bottom	26.0mm	26.0mm	Exact match
A button centre from right	22.0mm	22.0mm	Exact match
B button centre from right	32.0mm	32.0mm	Exact match
L shoulder length	52.0mm	52.0mm	Exact match
R shoulder length	52.0mm	52.0mm	Exact match
Shoulder button width	10.0mm	10.0mm	Exact match
USB-C port centre from bottom	N/A (original: EXT connector)	8.0mm	Bottom edge centre
3.5mm jack from right	N/A (original: top edge)	12.0mm from right, bottom edge	Relocated to bottom edge
MicroSD slot from right	N/A (original: link cable)	5.0mm from right edge, top third	Right edge access

5.2 Fusion 360 Workflow

Step	Fusion 360 Action	Detail
1	Create new component: GBA_Pro_Shell	All geometry lives inside this component
2	Sketch master outline on XY plane	144x82mm rectangle. Add corner radii: 12mm top, 8mm bottom. Fully constrain with dimensions.
3	Extrude to 25mm	This is your solid Master Body. Direction: symmetric, 12.5mm each way.
4	Create offset plane at Z=0.5mm	This will be the split line between front and back shell. Slightly above midplane so back is deeper.
5	Split Body	Modify → Split Body. Select Master Body. Splitting tool: offset plane. You now have Front Shell and Back Shell bodies.
6	Shell Front Body	Modify → Shell. Select Front Shell. Remove face: the flat back face. Wall thickness: 2.0mm. This hollows the front shell to 2mm walls.
7	Shell Back Body	Modify → Shell. Select Back Shell. Remove face: the flat front face. Wall thickness: 2.0mm.
8	Add display window cutout	New sketch on front face. Rectangle: 62x42mm, positioned 14mm from top, 41mm from left. Extrude-cut through all.
9	Add button wells (D-pad)	Sketch on front face. Cross shape: 32mm span, 10.5mm arm width. Extrude-cut 1.5mm deep (dome sits in this recess). Leave 0.5mm floor for dome retention.
10	Add button wells (A, B)	Two circles: 12mm diameter each. A: 22mm from right, 26mm from bottom. B: 32mm from right, 32mm from bottom. Extrude-cut 1.5mm deep.
11	Add Start/Select wells	Two rectangles: 22x8mm, 4mm apart, centred horizontally. Extrude-cut 1.5mm deep.
12	Add L/R shoulder button cutouts	In top face: 52x10mm slots at each end. Extrude-cut through shell thickness. Shoulder buttons are separate printed parts that snap in from top.
13	Add PCB standoffs (back shell)	Inside back shell: 4 cylindrical bosses at PCB corner positions + 1 centre. Boss diameter 6mm, height 3mm from floor. Drill M2 insert hole: 3.2mm diameter, 3mm deep.
14	Add display bezel ledge (front shell)	Inside front shell around display window: 1.5mm wide shelf, 1.5mm deep. Display glass rests on this shelf.
15	Add battery bay (back shell)	Recess in back shell floor: 101x56mm, 6mm deep, centred between PCB standoffs and back wall.
16	Add USB-C cutout (back shell edge)	On bottom edge: 10x4mm slot centred, 8mm from bottom. USB-C port emerges here.
17	Add 3.5mm jack cutout	Bottom edge: 7mm circle, 12mm from right edge.
18	Add MicroSD cutout	Right edge: 15x2.5mm slot, positioned at top third of right edge.
19	Add shell alignment pins	Front shell: 2x cylindrical pins, 3mm diameter, 5mm tall, on inside face at diagonal corners. Back shell: matching 3.2mm holes. These align the two halves.
20	Add screw boss holes	4 corners of back shell: 4mm boss, M2 through-hole clearance (2.2mm diameter). Front shell matching positions: M2 heat-set insert holes (3.2mm diameter, 5mm deep).
21	Export STLs	Right-click each body → Save as Mesh → STL. Export: Front_Shell.stl, Back_Shell.stl, L_Button.stl, R_Button.stl

5.3 FDM Print Settings

Parameter	Front Shell	Back Shell	Buttons
Material	PETG	PETG	PETG

Parameter	Front Shell	Back Shell	Buttons
Layer height	0.15mm	0.15mm	0.12mm (finer)
Infill	40%	40%	60%
Infill pattern	Gyroid	Gyroid	Grid
Perimeters / walls	4	4	5
Support	Yes — for button wells	Yes — for USB-C slot	No
Support type	Tree (organic)	Normal	—
Print orientation	Outside face DOWN	Outside face DOWN	As designed
Estimated time	~8 hours	~6 hours	~1 hour combined
Nozzle temp (PETG)	235°C	235°C	235°C
Bed temp (PETG)	80°C	80°C	80°C

5.4 Post-Print Finishing

Stage	Process	Materials
1 — Support removal	Remove all supports with flush cutters and needle-nose pliers	Flush cutters, needle-nose pliers
2 — Test fit	Dry-fit PCB, battery, display BEFORE sanding. Mark any interference areas with pencil.	PCB, battery, display
3 — Sand exterior	Start with 120 grit to remove layer lines. Progress: 120→220→400→600. Wet-sand 400+ for best results.	Sandpaper 120/220/400/600, water
4 — Heat-set inserts	Heat soldering iron to 220°C. Press M2 inserts into boss holes slowly. Do not overheat — PETG softens at 80°C.	Soldering iron, M2 brass inserts
5 — Fill & prime	Apply automotive spot filler (e.g. Bondo) to any remaining surface defects. Sand flat. Apply 2x coats filler primer.	Spot filler, filler primer spray
6 — Final sand	Sand primed surface with 800 grit. Wipe clean with isopropyl alcohol.	800 grit, IPA
7 — Top coat	Apply 2-3 thin coats gloss spray paint in desired colour. Allow 15 min between coats. Optional: clear coat for protection.	Gloss spray paint, clear coat

Chapter 6

Phase 4 — Firmware Stack

The GBA Pro runs bare-metal C/C++ on the RP2350. No operating system. The firmware is built on the Raspberry Pi Pico SDK and uses a GBA emulator core ported to the RP2350. This chapter covers development environment setup, display driver, input handling, audio, and emulator integration.

6.1 Development Environment Setup

Install the following tools on your development machine (Linux/macOS recommended, WSL2 on Windows):

Tool	Version	Install Command	Purpose
Raspberry Pi Pico SDK	Latest	git clone https://github.com/raspberrypi/pico-sdk	Base SDK for RP2350
CMake	3.17+	sudo apt install cmake	Build system
ARM GCC toolchain	12+	sudo apt install gcc-arm-none-eabi	C/C++ compiler for RP2350
Python 3	3.8+	sudo apt install python3	SDK scripts
Ninja	Latest	sudo apt install ninja-build	Fast builds
GDB + OpenOCD	Latest	sudo apt install gdb-multiarch openocd	Debug via SWD
Visual Studio Code	Latest	Download from code.visualstudio.com	IDE (optional but recommended)
Pico SDK VSCode ext	Latest	Install from VSCode marketplace	CMake integration

6.2 Firmware Architecture

The firmware is structured as a single CMake project with these modules:

Module	File(s)	Responsibility
Main loop	main.c	Initialise all hardware, start emulator core, handle system events
Display driver	display_ili9341.c/.h	SPI0 init, framebuffer push at 60fps using DMA
Input driver	input.c/.h	Poll 10 GPIO pins, debounce, map to GBA button register
Audio driver	audio_i2s.c/.h	PIO I2S output, ring buffer, mix GBA audio output
Storage driver	storage_sd.c/.h	SPI1 SD init via FatFS, ROM load, save state read/write
Power driver	power.c/.h	I2C MAX17048 read, battery % display, low-battery warning
Emulator core	peanut_gb/ or mgba-pico/	GBA CPU + PPU + APU emulation
UI overlay	ui.c/.h	Battery %, ROM selection menu, save state picker

6.3 Recommended Emulator Core

The best-tested GBA emulator for RP2350/2040 is **Peanut-GB** (for GB/GBC) combined with **mGBA** ported to RP2040 by David Guillen Fandos. The mGBA-pico port runs GBA at full speed on RP2040 at 250MHz — the RP2350 at 150MHz default has more headroom.

Repository: <https://github.com/davidgfnet/gba-pico> — this is your primary emulator codebase.

6.4 Display Driver — ILI9341 at 60fps

The ILI9341 must run on SPI0 at the maximum supported clock. The RP2350 SPI peripheral supports up to 62.5MHz on SPI0. The ILI9341 datasheet specifies 10MHz minimum write cycle, so 62.5MHz is fine. Use DMA to push frames without blocking the CPU.

Key initialisation sequence for ILI9341:

```
// ILI9341 init sequence (subset – critical commands)
ili9341_cmd(0x01); // Software reset
sleep_ms(150);
ili9341_cmd(0x11); // Sleep out
sleep_ms(120);
ili9341_cmd(0x3A); // Pixel format
ili9341_data(0x55); // 16-bit colour (RGB565)
ili9341_cmd(0x36); // Memory access control
ili9341_data(0x48); // Row/col order, BGR
ili9341_cmd(0x29); // Display ON

// Set display window to full 320x240
ili9341_set_window(0, 0, 319, 239);

// DMA framebuffer push – call every frame
dma_channel_config cfg = dma_channel_get_default_config(dma_ch);
channel_config_set_transfer_data_size(&cfg, DMA_SIZE_16);
channel_config_set_dreq(&cfg, spi_get_dreq(SPI0, true));
dma_channel_configure(dma_ch, &cfg,
&spi;_get_hw(SPI0)->dr, // write to SPI data register
framebuffer, // read from framebuffer
320 * 240, // 76800 pixels
true); // start immediately
```

6.5 Input Handling

Poll all 10 button GPIOs every frame (every 16.7ms at 60fps). Map directly to GBA button register bits:

```
#define BTN_A (1 << 0)
#define BTN_B (1 << 1)
#define BTN_SEL (1 << 2)
#define BTN_STA (1 << 3)
#define BTN_RIGHT (1 << 4)
#define BTN_LEFT (1 << 5)
#define BTN_UP (1 << 6)
#define BTN_DOWN (1 << 7)
#define BTN_R (1 << 8)
#define BTN_L (1 << 9)

uint16_t read_buttons(void) {
```

```

uint16_t state = 0xFFFF; // all released (active low in GBA)
if (!gpio_get(GP_A)) state &= ~BTN_A;
if (!gpio_get(GP_B)) state &= ~BTN_B;
if (!gpio_get(GP_SEL)) state &= ~BTN_SEL;
if (!gpio_get(GP_STA)) state &= ~BTN_STA;
if (!gpio_get(GP_RIGHT)) state &= ~BTN_RIGHT;
if (!gpio_get(GP_LEFT)) state &= ~BTN_LEFT;
if (!gpio_get(GP_UP)) state &= ~BTN_UP;
if (!gpio_get(GP_DOWN)) state &= ~BTN_DOWN;
if (!gpio_get(GP_R)) state &= ~BTN_R;
if (!gpio_get(GP_L)) state &= ~BTN_L;
return state; // pass this directly to emulator core
}

```

6.6 I2S Audio via PIO

The RP2350 has no dedicated I2S peripheral, but the PIO (Programmable I/O) unit handles it perfectly. The Pico SDK includes a ready-made I2S PIO program.

```

// audio_i2s.c - init
#include "hardware/pio.h"
#include "hardware/dma.h"
#include "audio_i2s.pio.h" // generated from .pio file

#define SAMPLE_RATE 32000
#define BCK_PIN 16
#define LRCK_PIN 17 // must be BCK_PIN + 1
#define DATA_PIN 18

void audio_init(void) {
    PIO pio = pio0;
    uint sm = pio_claim_unused_sm(pio, true);
    uint offset = pio_add_program(pio, &audio_i2s_program);
    audio_i2s_program_init(pio, sm, offset, DATA_PIN, BCK_PIN);
    // clock div for target sample rate
    float clk_div = (float)clock_get_hz(clk_sys)
        / (SAMPLE_RATE * 32 * 2);
    pio_sm_set_clkdiv(pio, sm, clk_div);
    pio_sm_set_enabled(pio, sm, true);
}

```

Chapter 7

Phase 5 — Fabrication & Assembly

This chapter covers ordering the PCB, assembling all components, printing the shell, and doing a first mechanical assembly. Do not connect power until the assembly checklist in Chapter 8 is complete.

7.1 PCB Ordering (JLCPCB)

Setting	Value
Board count	5 (minimum sensible quantity — keeps spares for rework)
Dimensions	130 x 70mm
Layers	2
PCB thickness	1.6mm
PCB colour	Black (personal preference — any colour works)
Silkscreen	White
Surface finish	ENIG (Electroless Nickel Immersion Gold) — REQUIRED for rubber dome contacts
Copper weight	1oz
Gold thickness	1U" (standard ENIG)
Via covering	Tented vias
Remove order number	Specify a location in silkscreen — JLCPCB adds their number here
Lead time	Standard: 5-7 business days + 3-5 days shipping
Estimated cost (5 boards)	~\$20–25 with ENIG

7.2 SMD Assembly Order

Assemble in this order. Larger components first, passive components last. Use solder paste and a heat gun or reflow station. Do NOT hand-solder QFN packages without hot air.

Order	Component	Package	Method	Critical Note
1	XC6220A LDO (U3)	SOT-25	Hot air / iron	Small — use flux, fine tip
2	TP4056 charger (U2)	SOP-8	Hot air / iron	PROG pin resistor R_PROG adjacent
3	MAX17048 fuel gauge (U4)	DFN-8	Hot air	No-lead package — paste on pad
4	PCM5102A DAC (U5)	TSSOP-20	Hot air / iron	0.65mm pitch — flux heavily
5	TPA2012D2 amp (U6)	QFN-16	Hot air + reflow	Thermal pad underneath — needs hot air
6	Pico 2 module (U1)	Stamp/castellated	Iron on castellated edges	Check every edge pin with continuity tester
7	All 0402 capacitors	0402	Iron or hot air	Use tweezers, toothpick for placement

Order	Component	Package	Method	Critical Note
8	All 0402 resistors	0402	Iron or hot air	Mark with marker before soldering
9	USB-C connector (J1)	Mid-mount	Iron	Check mechanical fit to shell before soldering
10	MicroSD slot (J3)	SMD through-post	Iron	Insert card and check alignment before final solder
11	3.5mm jack (J2)	PJ-320A SMD	Iron	Through-hole pins — easy to solder
12	Tact switches (SW10, SW11)	SMD tact	Iron	Shoulder button switches
13	Power switch (SW1)	Through-hole	Iron	Test action before soldering fully
14	Button dome PCB pads	PCB pads only	N/A — no soldering	ENIG pads are exposed — just clean with IPA
15	Battery connector (JST-PH)	Through-hole	Iron	Do NOT connect battery until power stage tested

CRITICAL: Do NOT connect the LiPo battery until you have verified 3.3V on the output rail with a multimeter. A wiring error on the power stage can destroy every component on the board.

7.3 Display Assembly

Step	Action
1	Test display before installation: connect via breadboard/wires to a spare Pico 2. Flash a display test sketch. Confirm pixels work.
2	Apply 0.5mm foam tape (3M 9448A recommended) around the display bezel — this cushions the glass against the shell.
3	Set display into shell front window from inside. It should sit on the 1.5mm bezel shelf you modelled in Fusion 360.
4	Route the FPC cable from display to the PCB FPC connector. Ensure no sharp bends in cable — minimum bend radius 3mm.
5	Do NOT glue the display yet — it stays removable via the front shell bezel ledge pressure fit.

7.4 Mechanical Assembly Sequence

Step	Action
1	Press M2 heat-set inserts into all shell boss holes using soldering iron at 220°C.
2	Install rubber dome sheet over button pad area on PCB. Align dome centres over pad centres. Secure with small piece of double-sided tape at one corner.
3	Install display into front shell (see 7.3). Route FPC cable through PCB slot.
4	Set PCB into back shell. Align with standoffs. Secure with 4x M2 x 5mm screws.
5	Connect FPC cable from display to PCB FPC connector. Lock connector latch.
6	Connect battery JST-PH to battery connector on PCB.
7	Route battery into battery bay. Press battery flat into bay — it should sit below PCB level.
8	Fit L and R shoulder button assemblies into their shell slots. Check snap action.
9	Bring front shell and back shell together. Align with alignment pins. Check no cables are pinched.
10	Secure with 4x M2 x 5mm screws through back shell into front shell inserts. Do not overtighten — PETG will strip.

Step	Action
11	Install button caps over dome wells (if using separate ABS button caps — optional, domes work exposed).

Chapter 8

Phase 6 — Bring-Up & Testing

Bring-up is the process of powering up a new PCB for the first time and systematically verifying each subsystem. Follow this sequence exactly. Each stage must pass before proceeding to the next. Never power the full board until Stage 1 passes.

8.1 Pre-Power Checklist

Check	How to verify	Pass condition
Visual inspection	Magnifier or phone camera at 2x. Check every solder joint.	No bridges, no cold joints, no tombstoned 0402s
Short circuit check — 3V3 rail	Multimeter continuity mode between 3V3 and GND	No continuity (no beep)
Short circuit check — VBAT	Multimeter continuity mode between VBAT and GND	No continuity
Short circuit check — USB 5V	Multimeter continuity mode between VBUS and GND	No continuity
USB-C orientation	Visual — CC1 and CC2 pads, 5.1k resistors present	Both CC pins have pull-down resistors
Pico 2 orientation	Visual — pin 1 marker aligned	Correct orientation
Capacitor polarity	Visual — electrolytic caps (if any) correct polarity	All polarised caps correct — 0402 ceramics are non-polar
Battery not connected	Visual	JST battery connector is NOT plugged in for Stage 1

8.2 Stage 1 — Power Stage Test (USB powered, no battery)

Test	Expected result	If fail
Connect USB-C to computer (no firmware yet)	No smoke, no heat	Disconnect immediately. Recheck for shorts.
Measure USB-C VBUS pin	$5.0V \pm 0.25V$	Check USB-C wiring
Measure TP4056 VIN pin	$5.0V \pm 0.25V$	Check trace from USB-C to TP4056
Measure XC6220 VIN (from VBAT net)	< 1V with no battery	Normal — no battery connected
Connect current-limited bench supply to JST instead of battery: 3.7V, 200mA limit	—	—
Measure XC6220 VOUT	$3.30V \pm 0.05V$	Check LDO part placement and input cap
Measure 3V3 on Pico 2 3V3 pin	$3.30V \pm 0.05V$	Check trace from LDO to Pico 2
LDO temperature after 30 seconds	Warm but not hot (< 50°C touch)	Check for unexpected current draw

8.3 Stage 2 — Firmware Flash & Basic Boot

Step	Action	Expected
1	Hold BOOTSEL on Pico 2 while connecting USB	RP2350 appears as USB mass storage drive (RPI-RP2)
2	Copy a simple LED blink UF2 (from Pico SDK examples) to the drive	Drive ejects, LED blinks at 1Hz on GP25
3	Flash display test firmware	ILI9341 shows colour gradient or test pattern
4	Flash button test firmware	Serial monitor shows button names as pressed
5	Flash audio test firmware	1kHz tone plays from speaker
6	Flash SD test firmware with FAT32 formatted card inserted	Serial shows card detected, file listing
7	Flash MAX17048 test firmware	Serial shows battery percentage reading
8	All stages passing: flash full emulator firmware	ROM selection menu appears on display

8.4 Stage 3 — Full System Test

Test	Method	Pass
Load a GBA ROM from SD card	Select from menu, press A to load	Game boots and runs at full speed
All buttons functional	Open a game's button test, press each button	All 10 inputs register cleanly
D-pad feel	Play a platformer for 5 minutes	No missed inputs, no double-triggers
Display quality	Check for dead pixels, backlight uniformity	No dead pixels, even backlight
Audio quality	Load a game with music	Clean audio, no crackling, no hiss
Battery charging	Connect USB-C with battery installed	TP4056 CHRG LED lights, MAX17048 shows rising %
Battery runtime	Fully charge, play until shutdown	> 6 hours gameplay
USB-C disconnect hot-swap	Disconnect USB while on battery	Seamless, no reset
Save state	Save and load a save state	Game state restored correctly
Physical fit	Check all buttons, no shell flex, no rattles	Rigid shell, all buttons travel correctly

Chapter 9

Troubleshooting Reference

Symptom	Most Likely Cause	Fix
No 3.3V output from LDO	LDO orientation reversed or bridged pins	Recheck XC6220A SOT-25 pin 1 orientation. Reflow.
3.3V present but low (< 3.0V)	LDO output cap missing or excessive load	Check 1uF output cap. Check for short on 3V3 rail.
Pico 2 not detected by PC as USB	No power to Pico 2, or BOOTSEL not held	Verify 3.3V on Pico 2 VSYS. Try holding BOOTSEL.
Display blank / all white	SPI not initialised, or CS/DC wiring error	Check GP5 (CS) is pulled low during init. Check DC pin logic.
Display garbled / wrong colours	Wrong SPI mode or colour format	ILI9341 uses SPI mode 0 (CPOL=0, CPHA=0). Pixel format must be RGB565 (0x55).
Display very slow (< 10fps)	SPI clock too low, or no DMA	Set SPI clock to 62.5MHz. Use DMA for frame push.
Buttons not registering	Pull-up resistors missing or wrong GPIO	Verify 10k pull-up on each button GPIO. Check GPIO assignment matches firmware.
Buttons double-triggering	No debounce in firmware, or dome seated wrong	Add 5ms debounce in firmware. Check dome alignment over pads.
No audio from speaker	I2S PIO not started, or amp in shutdown	Check TPA2012 SDZ pin is HIGH (not in shutdown). Check PIO clock div.
Audio crackling / distortion	Audio buffer underrun, or amp gain too high	Increase audio buffer size. Reduce gain setting on TPA2012.
SD card not detected	SPI1 wiring error or card not FAT32	Check GP12 (MISO) has pull-up. Format card as FAT32.
SD card detected but ROM won't load	File system corruption or filename too long	Use 8.3 filenames on SD. Reformat if needed.
Battery not charging	TP4056 PROG resistor wrong value	1.2k on PROG = 1A charge. Check resistor value.
Battery % not updating	MAX17048 I2C address wrong or no pull-ups	MAX17048 I2C address is 0x36. Add 4.7k pull-ups on SDA/SCL.
Shell halves don't align	Alignment pins wrong dimensions	Pins: 3mm diameter. Holes: 3.2mm. Check Fusion 360 model.
Rubber domes feel clicky not smooth	Dome seated incorrectly or pad too small	Ensure dome carbon face contacts both pad fingers. Pad diameter must be 7mm.
GBA runs slow (< 60fps)	CPU not overclocked, or emulator not optimised	Overclock RP2350 to 250MHz in firmware. Use mGBA-pico optimised build.
Screen tears during gameplay	No vsync between display DMA and emulator	Sync DMA frame start to display VSYNC signal or use double buffering.

Appendix A

Complete Pin Assignment Table

Complete mapping of every RP2350 / Pico 2 pin used in this design:

GPIO	Pico Pin	Signal Name	Connected To	Direction	Notes
GP0	1	BTN_RIGHT	D-pad Right dome pad	IN	10k pull-up to 3V3
GP1	2	BTN_LEFT	D-pad Left dome pad	IN	10k pull-up to 3V3
GP2	4	SPI0_CLK	ILI9341 SCK	OUT	SPI0 peripheral, 62.5MHz
GP3	5	SPI0_TX	ILI9341 MOSI/SDA	OUT	SPI0 peripheral
GP4	6	SPI0_RX	NC (display write-only)	IN	Not connected
GP5	7	SPI0_CSn	ILI9341 CS	OUT	SPI0 chip select, active low
GP6	9	LCD_DC	ILI9341 DC/RS	OUT	Data/command select
GP7	10	LCD_RST	ILI9341 RESET	OUT	Active low reset
GP8	11	LCD_BL	ILI9341 LED via NPN	OUT	PWM brightness control
GP9	12	BTN_R	R shoulder tact switch	IN	10k pull-up to 3V3
GP10	14	SPI1_CLK	MicroSD CLK	OUT	SPI1 peripheral, 25MHz
GP11	15	SPI1_TX	MicroSD CMD/MOSI	OUT	SPI1 peripheral
GP12	16	SPI1_RX	MicroSD DAT0/MISO	IN	SPI1 peripheral, 10k pull-up
GP13	17	SPI1_CSn	MicroSD CS	OUT	SPI1 chip select, active low
GP14	19	SD_DET	MicroSD card detect	IN	10k pull-up, low = card present
GP15	20	BTN_SEL	Select dome pad	IN	10k pull-up to 3V3
GP16	21	I2S_BCK	PCM5102A BCK	OUT	PIO I2S bit clock
GP17	22	I2S_LRCK	PCM5102A LRCK	OUT	PIO I2S word clock
GP18	24	I2S_DATA	PCM5102A DIN	OUT	PIO I2S data
GP19	25	BTN_STA	Start dome pad	IN	10k pull-up to 3V3
GP20	26	I2C1_SDA	MAX17048 SDA	I/O	4.7k pull-up to 3V3
GP21	27	I2C1_SCL	MAX17048 SCL	OUT	4.7k pull-up to 3V3
GP22	29	BTN_UP	D-pad Up dome pad	IN	10k pull-up to 3V3
GP26	31	BTN_DOWN	D-pad Down dome pad	IN	10k pull-up to 3V3
GP27	32	BTN_A	A button dome pad	IN	10k pull-up to 3V3
GP28	34	BTN_B	B button dome pad	IN	10k pull-up to 3V3
VSYS	39	3V3_IN	XC6220A VOUT (3.3V)	PWR	Main power input to Pico 2
GND	38	GND	Common ground	PWR	Multiple GND pins on Pico 2
USB_DP	—	USB_DP	USB-C D+ via 27-ohm	I/O	For firmware flashing only

GPIO	Pico Pin	Signal Name	Connected To	Direction	Notes
USB_DM	—	USB_DM	USB-C D- via 27-ohm	I/O	For firmware flashing only

Appendix B

Fusion 360 Dimension Master Sheet

Every Fusion 360 sketch dimension in one place. Use this as your modelling checklist:

Sketch / Feature	Dimension	Value	Constraint type
Master outline width	Horizontal	144.0mm	Fixed
Master outline height	Vertical	82.0mm	Fixed
Top corner radius	Radius	12.0mm	Fixed
Bottom corner radius	Radius	8.0mm	Fixed
Extrusion depth (total)	Symmetric	25.0mm	Fixed
Split plane offset from mid	Z offset	+0.5mm toward back	Fixed
Shell wall thickness	Uniform	2.0mm	Fixed
Display window width	Horizontal	62.0mm	Fixed
Display window height	Vertical	42.0mm	Fixed
Display window from top (inner)	Vertical	14.0mm	Fixed
Display window from left (inner)	Horizontal	41.0mm	Fixed
Display bezel shelf width	Uniform	1.5mm	Fixed
Display bezel shelf depth	Depth	1.5mm	Fixed
D-pad cross span	Diagonal tip to tip	32.0mm	Fixed
D-pad arm width	Width	10.5mm	Fixed
D-pad well depth	Depth	1.5mm	Fixed
D-pad centre from left (outer)	Horizontal	19.0mm	Fixed
D-pad centre from bottom (outer)	Vertical	26.0mm	Fixed
A button diameter (well)	Diameter	12.0mm	Fixed
B button diameter (well)	Diameter	12.0mm	Fixed
A button centre from right (outer)	Horizontal	22.0mm	Fixed
B button centre from right (outer)	Horizontal	32.0mm	Fixed
Button well depth	Depth	1.5mm	Fixed
L shoulder slot length	Length	52.0mm	Fixed
L/R shoulder slot width	Width	10.0mm	Fixed
PCB standoff height (back shell)	Height from floor	3.0mm	Fixed
PCB standoff outer diameter	Diameter	6.0mm	Fixed
M2 insert hole diameter	Diameter	3.2mm	Fixed
M2 insert hole depth	Depth	3.0mm	Fixed

Sketch / Feature	Dimension	Value	Constraint type
Battery bay length	Length	101.0mm	Fixed
Battery bay width	Width	56.0mm	Fixed
Battery bay depth	Depth	6.0mm	Fixed
USB-C slot width	Width	10.0mm	Fixed
USB-C slot height	Height	4.0mm	Fixed
USB-C slot from bottom edge	Vertical	8.0mm	Fixed
3.5mm jack hole diameter	Diameter	7.0mm	Fixed
3.5mm jack from right edge	Horizontal	12.0mm	Fixed
MicroSD slot width	Width	15.0mm	Fixed
MicroSD slot height	Height	2.5mm	Fixed
Alignment pin diameter (front)	Diameter	3.0mm	Fixed
Alignment hole diameter (back)	Diameter	3.2mm	+0.2mm clearance
Screw boss through-hole	Diameter	2.2mm	M2 clearance
Front shell insert hole	Diameter	3.2mm	M2 heat-set insert
Front shell insert depth	Depth	5.0mm	Fixed

Appendix C

Supplier & Ordering Reference

Supplier	URL	Best for	Shipping
Digikey	digikey.com	ICs, connectors, precision passives	Fast, expensive, reliable stock
Mouser	mouser.com	Same as Digikey, often better pricing on ICs	Fast, reliable
LCSC	lcsc.com	Cheap passives, Chinese ICs, bulk 0402s	2 weeks, very cheap
JLCPCB	jlcpcb.com	PCB fabrication	7–10 days from order to door
Adafruit	adafruit.com	Displays, Pico 2, sensors, LiPo cells	US-fast, international OK
Aliexpress	aliexpress.com	LiPo batteries, speakers, dome sheets	2–4 weeks, cheap
Handheld Legend	handheldlegend.com	GBA-spec rubber dome sheets, buttons	Specialist — worth it
Raspberry Pi	raspberrypi.com	Pico 2 direct purchase	Authorised resellers also
Amazon	amazon.com	M2 inserts, screws, sandpaper, PETG filament	Fast for consumables
Bambu Lab / Prusa	bambulab.com / prusa3d.com	PETG filament (if printing yourself)	Or local maker space

Recommended Order Sequence

Order	What to order	From	Why first
1st	Pico 2 + display + LiPo + rubber dome sheet	Adafruit + Handheld Legend	These take longest / most critical to test first
2nd	All ICs (TP4056, XC6220, MAX17048, PCM5102A, TPA2012)	Digikey / LCSC	Order together to consolidate shipping
3rd	All passives (0402 caps/resistors), connectors, switches	LCSC (bulk, cheap)	Order 50x of each 0402 value
4th	PCB fabrication	JLCPCB	Upload Gerbers after schematic DRC passes
5th	PETG filament, M2 inserts, screws, sandpaper	Amazon / local	Order while waiting for PCBs
6th	Speaker, headphone jack	Mouser / Aliexpress	Last — not needed until audio test stage

End of Document — GBA Pro Complete Engineering Reference Rev 1.0

Built with intention. Soldered with care. Played with nostalgia.